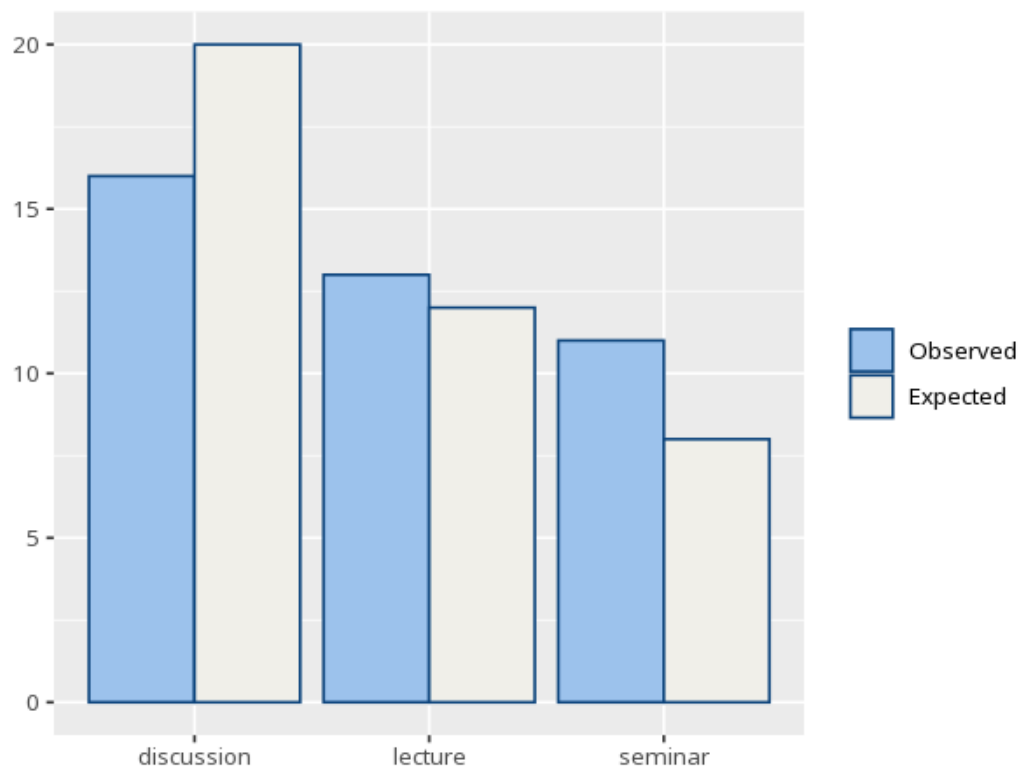


1 Chi-square goodness-of-fit

Category	Observed	Expected	Std. residual
discussion	16	20.00	-1.26
lecture	13	12.00	0.35
seminar	11	8.00	1.19

χ^2	df (k-1)	p	Cohen's w [95% CI]
2.01	2	.366	0.22 [0.00, 2.00]



Tests whether one categorical variable's counts depart from an expected distribution (equal by default). A p below alpha means the observed split differs from expected; a standardized residual beyond about ± 1.96 flags a category that significantly drives the result. Valid only when expected counts are mostly ≥ 5 (some texts allow ≥ 1 if no more than 20% are below 5); for sparse categories use an exact / simulated test instead. Your significance threshold (α) is 0.05.

A goodness-of-fit test, $\chi^2(2, N=40)=2.01$, $p = .366$, $w=.22$ [.00, 2.00].

Statistical basis: Chi-square goodness-of-fit test (Pearson, K. (1900). "On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling." Philosophical Magazine, 50(302), 157-175.);

Cohen's w effect-size benchmark (Cohen, J. (1988). *Statistical Power Analysis for the Behavioral Sciences* (2nd ed.). Routledge.).